

Understanding Tensors

Module 11 Study Notes • Machine Learning Data Structures

1. What is a Tensor?

To perform Machine Learning (and Deep Learning), algorithms must be fed data. However, algorithms only understand numbers, not text, images, or audio. Therefore, all data must be converted into numerical formats and stored efficiently.

A **Tensor** is simply a generic mathematical container for storing these numbers. In computer science terminology, a Tensor is equivalent to an **N-dimensional array**. It is the fundamental data structure used by all major ML/DL libraries (like NumPy, TensorFlow, and PyTorch).

Core Principle: Higher-dimensional Tensors are created by grouping together lower-dimensional Tensors. For example, a 1D Tensor is a collection of 0D Tensors. A 2D Tensor is a collection of 1D Tensors, and so on.

2. Terminology: Rank, Axes, and Shape

Before diving into the types of Tensors, you must understand three key terms:

- **Axes (or Rank / Dimensions):** The number of directions you can move within the data structure. A standard spreadsheet table has 2 axes (Rows and Columns). The number of axes is also called the "Rank" or "Dimensions" (e.g., Rank 2 = 2D Tensor).
- **Shape:** A tuple indicating how many elements exist along each axis. For a matrix with 4 rows and 3 columns, the shape is **(4, 3)**.
- **Size:** The total number of numerical items inside the Tensor. It is calculated by multiplying all the numbers in the shape tuple. For shape **(4, 3)**, the size is 12.

3. Types of Tensors (0D to 5D)

A. 0D Tensor (Scalar)

A single, standalone number. It has 0 axes.

- *Example:* The number **7**.

- *Shape:* $()$

B. 1D Tensor (Vector)

An array or list of numbers. It is a collection of Scalars. It has 1 axis.

- *Example:* A single row of data for one student $[7.2, 110, 1]$ (representing CGPA, IQ, and Placement Status).
- *Shape Example:* $(3,)$

C. 2D Tensor (Matrix)

A grid of numbers, consisting of rows and columns. It is a collection of 1D Tensors (Vectors). It has 2 axes.

- *Example:* A standard CSV file or Excel spreadsheet containing data for 10,000 students across 3 columns.
- *Shape Example:* $(10000, 3)$

D. 3D Tensor

A cube of numbers. It is a collection of 2D Tensors (Matrices). It has 3 axes.

- *Example 1: Time Series Data.* Tracking the High/Low stock prices for a company over 365 days yields a 2D matrix of $(365, 2)$. If you collect this data for 10 different companies, you stack those matrices together, resulting in a 3D Tensor with shape $(10, 365, 2)$.
- *Example 2: NLP (Natural Language Processing).* Words are converted into vectors (e.g., a word becomes a 1D vector of 4 numbers). A sentence is a sequence of words (a 2D matrix). A paragraph or collection of sentences becomes a 3D Tensor.

E. 4D Tensor

A collection of 3D Tensors. It has 4 axes. This is heavily used in Computer Vision.

- *Example: Image Data.* A single colored image is actually a 3D Tensor. It has a height (in pixels), a width (in pixels), and 3 color channels (Red, Green, Blue). Therefore, one image has the shape $(Height, Width, 3)$.
- If you are training an ML algorithm on a batch of 50 colored images, you group them together into a 4D Tensor.

- *Shape Example:* **(50, 1080, 1920, 3)** → (Batch Size, Height, Width, Channels).

F. 5D Tensor

A collection of 4D Tensors. It has 5 axes.

- *Example: Video Data.* A video is simply a rapid sequence of images (frames). If 1 image is a 3D Tensor, a sequence of images (one video) is a 4D Tensor.
- If you want to train an algorithm on a batch of 4 different videos, you stack those 4D Tensors together into a 5D Tensor.
- *Shape Example:* **(4, 1800, 1080, 1920, 3)** → (Number of Videos, Total Frames per Video, Height, Width, Channels).

4. Summary Table

Dimensions / Rank	Mathematical Name	Composition	Common Real-World Example
0D	Scalar	A single number	A single data point (e.g., Age = 25)
1D	Vector	Array of Scalars	A single row in a dataset
2D	Matrix	Array of Vectors	A standard tabular dataset (CSV/ Excel)
3D	3D Tensor	Array of Matrices	Time-Series Data, NLP Sentences
4D	4D Tensor	Array of 3D Tensors	A batch of colored Images
5D	5D Tensor	Array of 4D Tensors	A batch of Videos